Global Economics Comment: Supply Shocks from the War in Iran Are Mostly Limited to Energy

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- We estimate that higher oil prices will lower global GDP by 0.3% and boost headline inflation by 0.5-0.6 over the next year (with a smaller 0.1-0.2pp boost to core inflation) based on our commodity team's updated oil forecasts. A similar surge in natural gas prices will lead to incremental price pressure and growth headwinds in Europe and Asia. Risks are skewed toward larger effects as long as the Strait of Hormuz remains closed.
- The recent selloff in front end rates signals an expectation that global central banks will be especially sensitive to inflation given the memory of post-pandemic supply disruptions and associated surge in inflation. While the size of the energy shock will likely prompt some caution (and we have accordingly delayed our forecast for cuts in several economies) a key difference from 2021-2022 is that the supply shock today appears narrowly concentrated in the energy sector. We highlight three reasons why impacts on non-energy supply will likely be less severe.
- First, non-energy trade with Gulf economies is limited, accounting for only 1% of global trade, with major DMs and EMs generally even less exposed.
- Second, although Gulf exporters account for a large share of global production for some chemical and metal products, it appears less likely that trade disruptions for these products will create severe bottlenecks that limit broader production. Current spot price increases for the most highly exposed chemicals and metals suggest only 0.1pp incremental upside to global headline inflation.
- Third, the problematic shipping disruptions from 2021-2022 appear less likely to repeat. Gulf economies are not major re-exporters of most products (aside from boats), non-tanker ocean shipping costs have actually ticked down since the start of the war, and the recent rise in airfreight prices (while notable) implies less than 5bp upside to global inflation.
- Taken together, our analysis suggests that the major risk to global supply and inflation is mostly confined to energy, which limits the risk that the large second-round inflation effects observed in 2021-2022 will re-emerge.

Supply Shocks from the War in Iran Are Mostly Limited to Energy

Our commodity team's recent oil and gas price forecast upgrades imply that the rise in energy prices due to the closure of the Strait of Hormuz will have a larger impact

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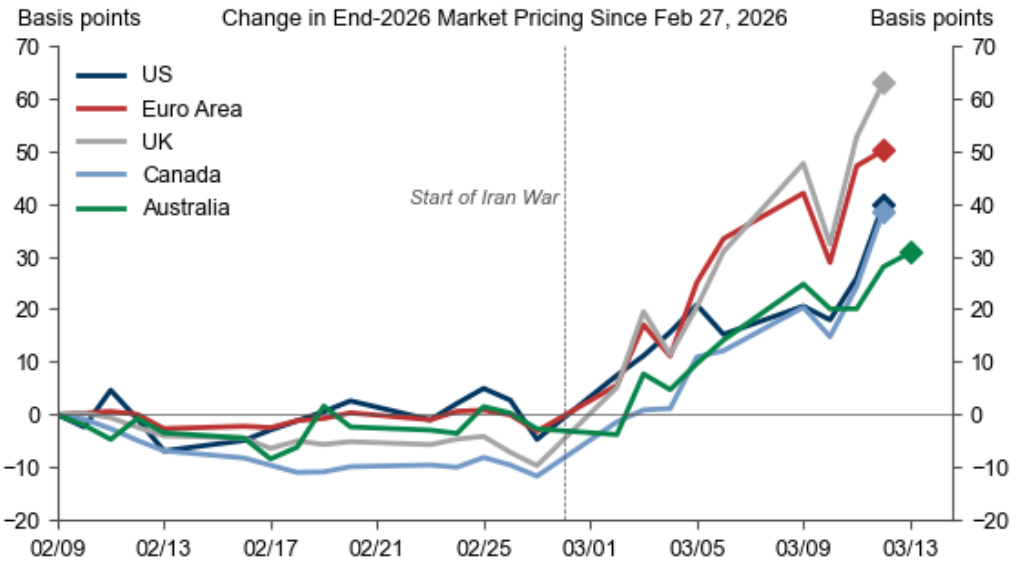
on the global economy. Our standard rules of thumb imply that our new energy price forecasts will lower global GDP by 0.3% and boost headline inflation by 0.5-0.6pp over the next year (with a smaller 0.1-0.2pp boost to core inflation). Following forecast changes in the US, Europe, Asia, LatAm, and CEEMEA, we now accordingly forecast 2.6% global growth (vs. 2.9% prior to the war) on an annual average basis and 2.9% headline inflation on a Q4/Q4 basis (vs. 2.3%).

The recent selloff in front end rates—Exhibit 1 shows that end-2026 policy rate pricing has increased notably in the UK (+63bp), Euro Area (+50bp), US (+40bp), Canada (+39bp), and Australia (+31bp) since the start of the war—signals an expectation that global central banks will be especially sensitive to higher inflation given the recent memory of post-pandemic supply disruptions and associated surge in inflation.

Exhibit 1: Front-End Rates Have Sold Off Since the Start of the War in Iran



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

The size of the energy shock will likely prompt some caution, and we have accordingly delayed our forecast for cuts in several economies, including the US and UK. That said, the biggest policy risk from higher energy prices is that second-round inflation spillovers could lead to sustained upward pressure on underlying inflation, as occurred in 2021-2022. A key difference between 2021-2022 and today, however, is that today's shock is more narrowly concentrated in the energy sector, whereas the energy price increases in 2022 were only one aspect of a much broader global supply chain crisis and inflation surge.

In this *Global Economics Comment* we highlight three reasons why impacts on non-energy supply will likely be less severe.

First, most major economies have limited trade exposure to the Middle East. While some regional economies are highly exposed—our Middle Eastern economics team sees significant headwinds to regional growth—non-energy exports account for less than 1% of total imports for major DMs and EMs, and only 1% of total global trade. As a point of comparison, the post-pandemic shutdowns to China and East Asian trade affected over 20% of global trade, suggesting that supply chain disruptions from the war in Iran will be much more limited than those after the pandemic.

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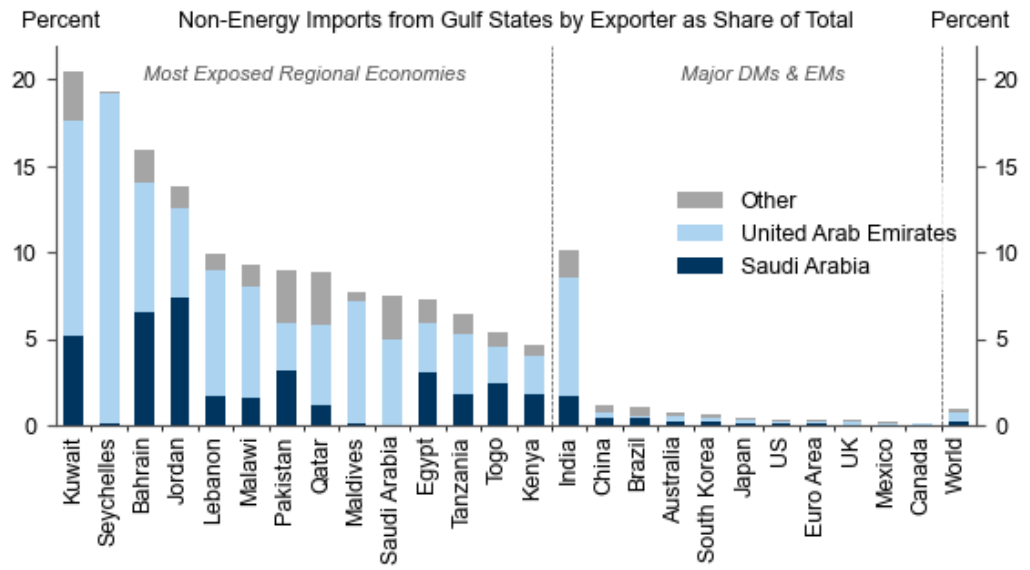
Exhibit 2: Outside of Regional Economies and India, Major DM and EM Economies Have Limited Trade Exposure to Non-Energy Goods

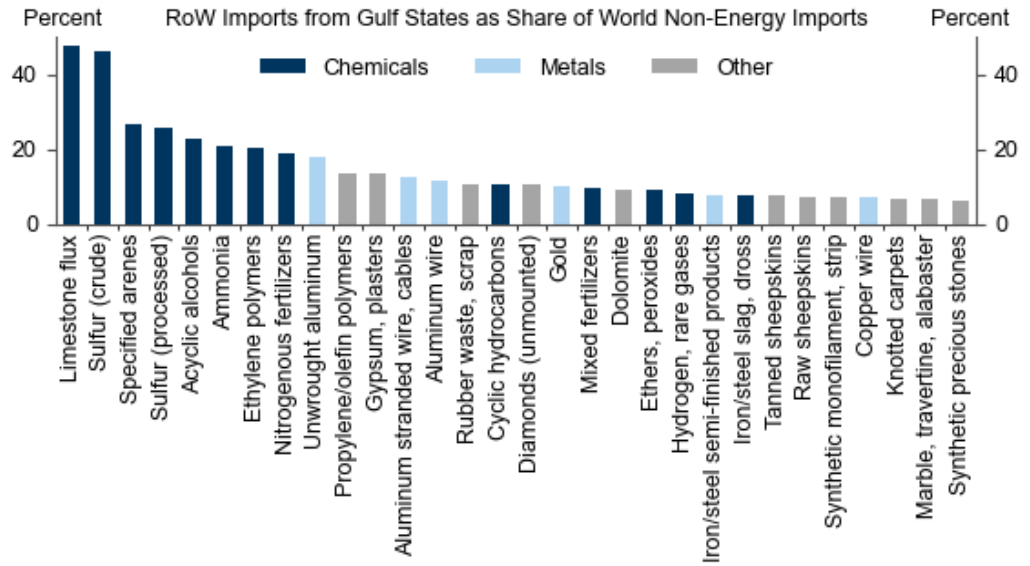


Source: UN Comtrade Database, Goldman Sachs Global Investment Research

Second, the production bottlenecks that emerged in 2021-2022 due to loss of supply of critical inputs and "just-in-time" inventory management seem less likely to repeat.

To identify key supply risks, in Exhibit 3 we examine Middle Eastern export shares of HS4 trade categories to assess where chokepoints are likely to emerge. The takeaway from this calculation is that Middle Eastern exporters account for a large share of global exports for some chemical and metal products that would likely be the source of any production bottlenecks.

Exhibit 3: Gulf Exporters Account for a Large Share of Global Trade for Some Chemical and Metal Products



Source: UN Comtrade Database, Goldman Sachs Global Investment Research

Exhibit 4 provides more context around potential bottlenecks. The left chart shows that Gulf states account for a large share of global production (in addition to the large shares

of trade shown in Exhibit 3) for several of these items. But the right chart shows that most of these highly exposed chemicals and metals account for at most 0.2% of global GDP, with Middle East production accounting for less than 0.02% of global GDP. In contrast, disruptions to global semiconductor production in 2022 affected trade worth around 1% of global GDP.

More importantly, these products do not appear likely to create shortages in "critical" inputs that would lead to broad production shutdowns. Sulfur, nitrogen, and ammonia are used as fertilizers, which increases agricultural productivity but is not essential and could be rationed. Most limestone flux is produced domestically, and polyethylene is mostly used for production of low value add (and generally not essential) plastics. We therefore anticipate limited production challenges from these items.

On the surface, helium would seem like the most likely candidate for potential supply disruptions given that is a hard-to-substitute input into production of MRI machines, semiconductors, and aerospace systems. But conversations with our equity analysts suggest that contracts for helium supply are generally long-term and fixed, and several large producers have enough helium stored to offset potential supply disruptions for the foreseeable future.

The most likely candidate source for production disruption is therefore methanol, which is a key input into production of acetic acid, which is in turn used to produce many industrial adhesives, solvents, and paints. While we don't see clear chokepoints for now, the potentially permanent loss of Iran's almost 20% of global production capacity raise some risk of downstream ripple effects.

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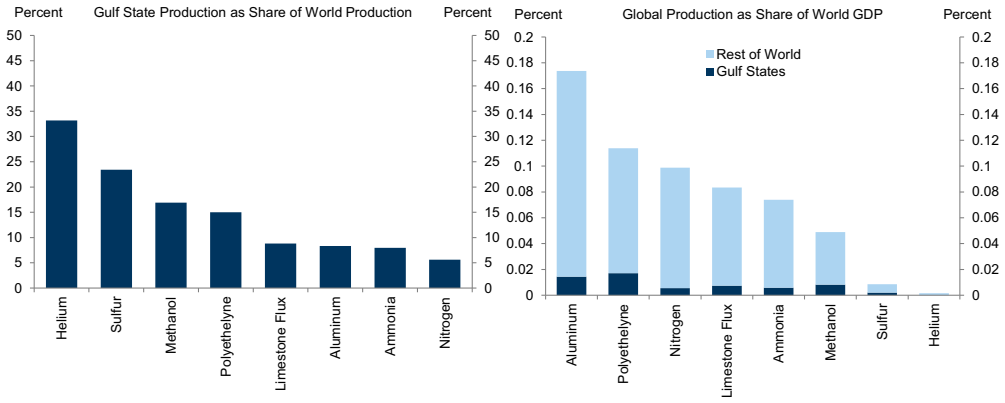
Exhibit 4: Gulf Countries Account for a Large Share of Global Production for Some Chemicals and Metals, but These Are Small as a Share of Global GDP



Source: Goldman Sachs Global Investment Research, Data compiled by Goldman Sachs Global Investment Research

Even if supply disruptions are not binding constraints on downstream production, the loss of supply is putting upward pressure on prices that could boost inflation. In Exhibit 5 we assess these risks by combining price increases so far with input shares to estimate the impact on global inflation.

The key challenge in analyzing the inflation effect is that most affected commodities' prices are not quoted regularly. The left chart of Exhibit 5 addresses this issue by compiling reported spot price increases for each product from available price indices and industry reports. The upshot of this exercise is that prices have increased by around

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25% for the considered products.

The right chart of Exhibit 5 combines these reported price increases with our estimates of global production (as a share of global GDP) to estimate the global inflation boost if consumers bear 70% of the cost. Our estimates suggest up to a 0.1pp boost to global headline inflation over the next 12 months if prices stay at current spot price levels, although the impact will be much smaller if spot prices normalize with supply following the reopening of the Strait of Hormuz.

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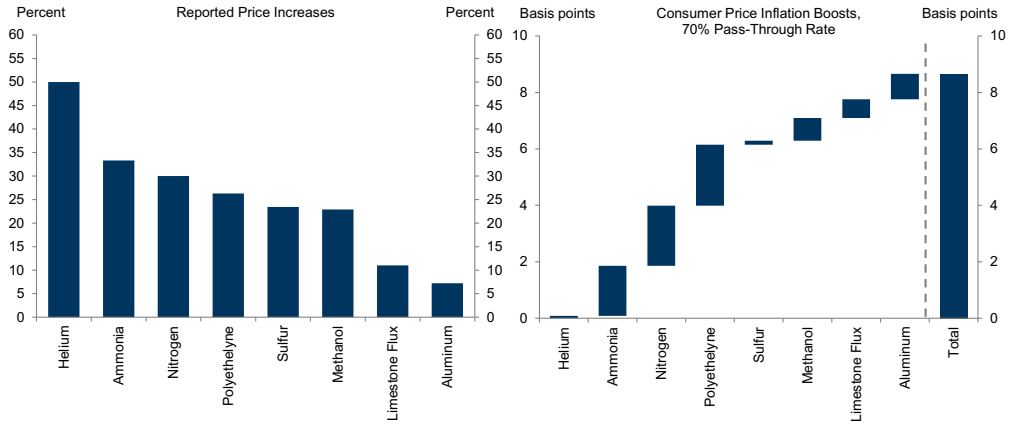
Exhibit 5: Prices Have Increased Significantly for Highly Exposed Chemical and Metals, but the Impact on Inflation Appears Limited, even if Prices Remain at Current Levels



Source: Goldman Sachs Global Investment Research, Data compiled by Goldman Sachs Global Investment Research

Third, the problematic shipping disruptions from 2021-2022—where port bottlenecks made it difficult to obtain supply even after production had recovered, leading some companies with inadequate inventory buffers to cut production—appear less likely to repeat. There are two channels through which disrupted trade flows could impact supply and prices.

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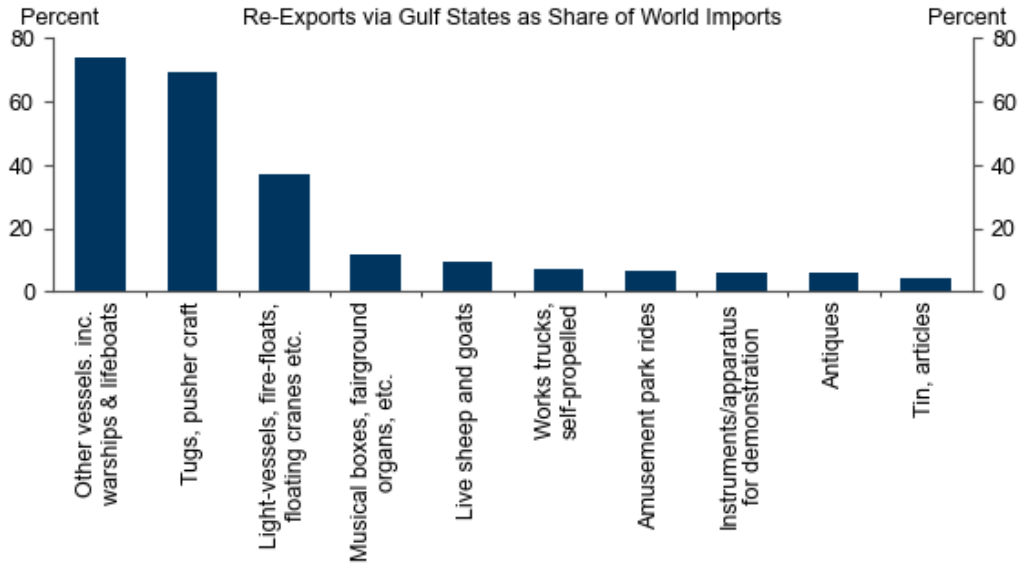
First, if the Middle East were a significant trade hub (that collected products and reallocated them to their destination), then a disruption in re-exports could lead to supply shortages. Exhibit 6 shows that trade rerouting could negatively affect re-exports of yachts and other boats, but the impact on other goods appears limited.

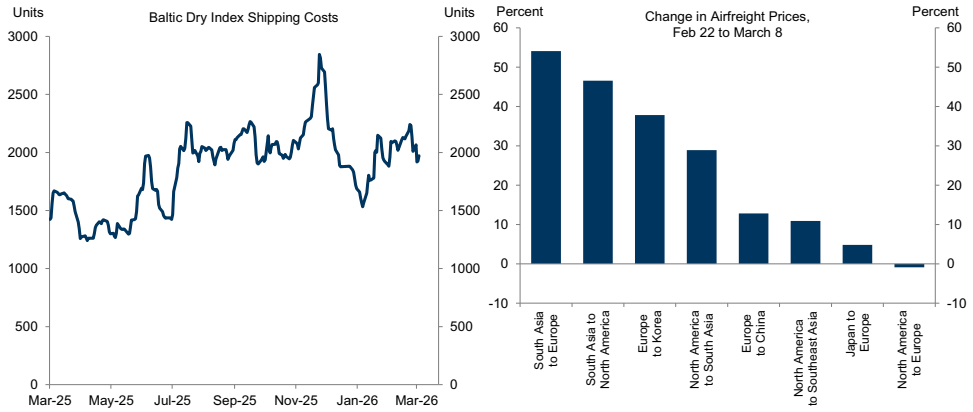
Exhibit 6: Gulf Country Reexports Are Limited Outside of Yachts



Source: UN Comtrade Database, Goldman Sachs Global Investment Research

Second, a reshuffling of global supply could lead to upward pressure on shipping prices. For ocean freight—which accounts for 90% of global shipping volume—this does not appear to be an issue thus far, as shipping prices ticked down last week according to the Baltic Dry Freight Index. In contrast, airfreight prices have increased significantly due to Middle Eastern air traffic disruptions, with cost increases to and from Asia and North America/Europe increasing by an average of 25%. That said, these spot price increases point to only 3bp upside to global inflation (with smaller impacts if costs normalize from current levels).

Exhibit 7: Air Freight Prices Have Increased, but Ocean Shipping Costs Have Ticked Down



Source: Goldman Sachs Global Investment Research, Bloomberg, Data compiled by Goldman Sachs Global Investment Research

In summary, our analysis suggests that the major risk to global supply and inflation is mostly confined to energy, which limits the risk that the severe supply chain disruptions

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(and associated surge in inflation) and large second-round inflation effects observed in 2021-2022 will re-emerge. This conclusion complements the findings from our European economics team that the economy is less exposed to energy shocks today than in 2022, although non-linear effects on energy cost passthrough and inflation expectations are possible if the energy crisis intensifies.

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